

Ch. 6 — P01 (K) Implement M/M/c in SimPy

Problem Bank | Chapter 6 | Type: K | Difficulty:

Chapter: 6 Type: K Difficulty:

Implement an M/M/c queue in SimPy and verify it against the Erlang-C formula.

System parameters:

- Arrival rate: $\lambda = 12$ customers/hour
- Service rate per server: $\mu = 5$ customers/hour
- Number of servers: $c = 3$
- Run for 5,000 customers per replication; delete warmup of 500

Tasks

1. **Implement:** Write a SimPy model `MMcQueue` that accepts λ , μ , c , `n_customers`, and `seed` as parameters. Use `simpy.Resource(env, capacity=c)`. Collect per-customer wait in queue W_q and sojourn W .
2. **Run 30 replications** using `np.random.default_rng(seed).spawn(30)`. Compute $\hat{W}_q$ (mean across replications) and a 95% CI.
3. **Erlang-C theory:** Compute the theoretical W_q using:

$$C(c, \rho) = \frac{\frac{(c\rho)^c}{c! (1 - \rho)} \cdot \frac{1}{c}}{\sum_{k=0}^{c-1} \frac{(c\rho)^k}{k!} + \frac{(c\rho)^c}{c! (1 - \rho)}} \quad W_q = \frac{C(c, \rho)}{\mu c - \lambda}$$

where $\rho = \lambda / (c\mu)$. Implement `erlang_c_wq(lam, mu, c)` in Python and compute the theoretical value.

4. **Comparison:** Does the 95% CI from Task 2 contain the theoretical value? Report the relative error $|\hat{W}_q - W_q^*| / W_q^*$.
5. **Sweep over c :** For $c \in \{2, 3, 4, 5\}$ and the same λ , μ , compute both the simulated $\hat{W}_q$ (10 reps each) and the theoretical W_q^* . Plot both on the same axis (theory as a line, simulation as points with error bars). At what c does W_q first fall below 1 minute?